

Words Beyond the Rate: High-Frequency Monetary Policy Shocks and FOMC Language

Abstract

Does FOMC statement language primarily reflect current monetary policy implementation or forward-looking informational revelation about future economic conditions? This paper examines this question using high-frequency monetary policy shocks derived from the Gürkaynak-Sack-Swanson (GSS) decomposition together with textual sentiment analysis of FOMC statements from 2006 - 2022.

The analysis shows that target shocks are more consistently associated with FOMC statement sentiment than path shocks, while path effects are generally weaker and sensitive to specification. Cross-asset evidence further indicates stronger responses among small-cap equities than large-cap equities, consistent with heterogeneous monetary transmission through financing conditions and discount-rate channels. Treasury and forward-guidance-related effects are comparatively limited in daily-frequency specifications.

Importantly, the paper does not interpret the GSS path factor as a pure information shock in the Jarociński-Karadi (2020) sense. Rather, the path factor captures revisions to expected future policy expectations that may reflect a combination of forward guidance, informational revelation, and expectation-related mechanisms.

Overall, the findings suggest that FOMC statement sentiment appears more closely tied to current policy implementation and financing conditions than to strong forward-looking informational revelation. More broadly, the paper contributes to the monetary policy communication literature by treating FOMC language as an endogenous output shaped by policy conditions rather than as a purely exogenous explanatory variable.

Keywords: Monetary policy, FOMC, Forward guidance, Sentiment analysis, High-frequency identification

JEL Classification: E43, E52, E58, G12, G14

1. Introduction

Central bank communication has become an increasingly important instrument of monetary policy. Over the past two decades, FOMC statements have evolved from brief post-meeting announcements into detailed policy communications designed to shape market expectations regarding inflation, economic conditions, and future interest-rate paths. Modern monetary policy therefore

operates not only through current policy actions, but also through communication itself.

Yet an important question remains unresolved:

Does FOMC statement language primarily reflect current policy implementation, or does it reveal forward-looking information about future economic conditions?

This distinction is central to the interpretation of monetary policy transmission. If FOMC language primarily reflects current policy implementation, then statement sentiment should be most closely associated with unexpected current-rate decisions. If, instead, FOMC communication primarily reveals information about future economic conditions, then revisions to expected future policy paths should play a more important role.

A large literature studies how financial markets respond to monetary policy announcements. High-frequency identification studies show that unexpected monetary policy shocks significantly affect equities, Treasury yields, and other financial assets (Kuttner, 2001; Gürkaynak, Sack, and Swanson, 2005a). More recent work emphasizes the information channel of monetary policy, arguing that central bank announcements may reveal information about future economic conditions (Campbell et al., 2012; Nakamura and Steinsson, 2018; Jarociński and Karadi, 2020). At the same time, an expanding communication literature applies textual analysis and sentiment methods to central bank language (Lucca and Trebbi, 2009; Apel and Blix, 2014).

However, relatively little attention has been devoted to a more fundamental question:

What determines the language itself?

Unlike most prior work, which studies how markets respond to FOMC communication, this paper studies what drives the language of the statements themselves. Rather than treating communication as a purely exogenous explanatory variable, the paper treats FOMC language as an endogenous output shaped by monetary policy conditions and policy surprises.

To study this question, the paper combines high-frequency monetary policy shocks with textual sentiment analysis of FOMC statements. Specifically, the analysis uses the Gürkaynak-Sack-Swanson (GSS) decomposition of monetary policy surprises into target shocks and path shocks. The target shock captures the unexpected component of the current policy-rate decision, while the path shock captures revisions to expected future policy paths.

Importantly, the GSS path factor should not be interpreted as a pure information shock in the Jarociński-Karadi (2020) sense. Rather, the path factor captures revisions to future policy expectations that may reflect forward guidance, informational revelation, or broader expectation-related mechanisms. This distinction is critical because the path factor may contain both commitment-related and information-related components.

The paper produces three main findings.

First, target shocks are more consistently associated with FOMC statement sentiment than path shocks. The target coefficient is statistically significant in baseline specifications, while the path coefficient is generally weaker and more sensitive to specification. Although formal coefficient-equality tests cannot statistically reject equal effects, the overall evidence provides more support for the policy-implementation interpretation than for a strong informational-revelation interpretation.

Second, cross-asset responses are strongest among equities, particularly small-cap equities. The equal-weighted CRSP index responds more strongly than the value-weighted index and the S&P 500, consistent with heterogeneous monetary transmission through financing conditions and balance-sheet sensitivity among smaller firms.

Third, evidence supporting a strong forward-guidance sentiment channel is comparatively limited in daily-frequency specifications. Treasury and forward-guidance-related effects are generally weaker and less stable than equity responses.

Taken together, the results suggest that FOMC statement sentiment appears more closely tied to current policy implementation and financing conditions than to strong forward-looking informational revelation.

This paper contributes to the literature in four ways. First, it studies what drives FOMC statement language itself rather than treating communication as purely exogenous. Second, it integrates high-frequency monetary policy shocks, textual sentiment analysis, and cross-asset evidence within a unified empirical framework. Third, it provides evidence consistent with heterogeneous monetary transmission across asset classes and firm sizes. Fourth, it highlights the importance of properly identified monetary policy surprises in monetary policy communication studies.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes the data and variable construction. Section 4 presents the empirical framework. Section 5 reports the main empirical results. Section 6 discusses robustness and extensions. Section 7 concludes.

2. Literature Review

2.1 Monetary Policy Surprises and High-Frequency Identification

The identification of exogenous monetary policy shocks has long been a central challenge in macroeconomics. Early approaches relied on narrative methods (Romer and Romer, 1989) or structural VARs (Christiano, Eichenbaum, and Evans, 1999), both of which depend on strong identifying assumptions regarding policy counterfactuals and economic timing relationships.

Kuttner (2001) introduced the high-frequency identification approach by using changes in federal funds futures around FOMC announcements to isolate the unexpected component of monetary policy decisions. The central insight is that only the unanticipated component of policy should affect asset prices contemporaneously. Kuttner shows that unexpected rate changes have economically significant effects on equities, while anticipated policy changes have comparatively limited effects.

Gürkaynak, Sack, and Swanson (2005a) extend this framework by decomposing monetary policy surprises into two dimensions: a target factor associated with the current policy-rate decision and a path factor associated with revisions to expected future policy paths. Their results demonstrate that future-policy revisions explain a substantial portion of long-term interest-rate movements and asset-price responses, highlighting the importance of forward guidance and expectations management in monetary transmission.

The GSS decomposition has since become a standard framework in empirical monetary economics. Swanson (2021) extends the target and path factors through 2019, while Acosta (2022) further extends the series through 2022, providing the shock data used in this paper.

More recently, Bauer and Swanson (2023) argue that high-frequency monetary policy surprises may be partially predictable using publicly available macroeconomic information released prior to FOMC meetings. This critique is important for interpretation because it implies that high-frequency surprises may contain endogenous information about economic conditions rather than purely exogenous policy innovations. Nevertheless, high-frequency identification remains one of the most widely used approaches for measuring monetary policy surprises because it substantially improves identification relative to simple realized rate changes.

2.2 Central Bank Communication

A growing literature examines how central bank communication affects financial markets and economic expectations. Blinder et al.

(2008) argue that communication has become an increasingly important policy instrument because modern monetary policy operates largely through expectation management.

Lucca and Trebbi (2009) apply computational text analysis to FOMC statements and show that statement complexity and language affect financial-market responses. Subsequent research expands the use of textual analysis in central banking contexts.

Apel and Blix (2014) construct a hawkish-dovish sentiment index for the Swedish Riksbank, demonstrating that central-bank-specific dictionaries can capture meaningful variation in monetary policy language. Hansen, McMahon, and Prat (2018) study transparency and deliberation within FOMC communications, while Rosa (2011) finds that statement tone affects Treasury yields and exchange rates.

Most importantly for this paper, the communication literature generally studies how markets respond to central bank language. By contrast, this paper studies what determines the language itself. Rather than treating communication as a purely exogenous explanatory variable, the paper treats FOMC statement language as an endogenous output shaped by monetary policy conditions and policy surprises.

2.3 The Information Channel of Monetary Policy

The information-channel literature argues that monetary policy announcements may reveal information about future economic conditions in addition to implementing policy decisions.

Romer and Romer (2000) provide early evidence that the Federal Reserve possesses informational advantages regarding inflation and macroeconomic conditions. Campbell et al. (2012) distinguish between “Delphic” forward guidance, which reveals information about the Federal Reserve’s economic outlook, and “Odyssean” forward guidance, which commits the central bank to a future policy path.

Nakamura and Steinsson (2018) show that monetary policy announcements significantly affect long-term expectations, which they interpret as evidence of informational effects. Jarociński and Karadi (2020) further separate monetary policy shocks from information shocks using sign restrictions, demonstrating that the two types of shocks generate opposite stock-market responses.

This paper relates to the information-channel literature but differs conceptually from the Jarociński-Karadi framework. Importantly, the GSS path factor should not be interpreted as a pure information shock. Rather, it captures revisions to expected future policy paths that may reflect informational revelation, forward-guidance commitment effects, or broader expectation-management mechanisms.

The paper therefore examines whether FOMC statement sentiment is more closely associated with current policy implementation (target shocks) or future-policy revisions (path shocks).

2.4 Sentiment Analysis and Central Bank Language

Sentiment analysis has become widely used in financial economics. The Loughran-McDonald (2011) dictionary remains the standard dictionary-based approach for financial text analysis, correcting several limitations associated with general-purpose sentiment dictionaries.

However, central bank communication presents unique challenges for sentiment analysis. FOMC statements are highly institutionalized documents that often maintain relatively positive or neutral language regardless of policy stance, creating positivity bias when standard financial dictionaries are applied to monetary-policy communication.

To address this issue, several studies construct central-bank-specific sentiment dictionaries designed to capture hawkish and dovish language more effectively (Apel and Blix, 2014; Tadler, 2022).

This paper adopts a combined approach that integrates a standard financial dictionary with a central-bank-specific sentiment dictionary. The objective is not to develop a new NLP methodology, but rather to use textual sentiment as a measurement tool for studying monetary policy communication.

More advanced contextual NLP approaches, including transformer-based models and large language models, may improve measurement precision in future research. However, the relatively small size and highly institutionalized structure of the FOMC statement corpus make transparent dictionary-based methods appropriate for the present analysis.

2.5 Forward Guidance and the Zero Lower Bound

The global financial crisis and zero lower bound period substantially increased the importance of forward guidance in monetary policy implementation. With short-term policy rates constrained near zero, the Federal Reserve increasingly relied on communication regarding the future path of interest rates as a policy instrument.

The forward-guidance period therefore provides a useful environment for examining whether statement sentiment becomes more strongly associated with future-policy-path revisions. However, the period also coincided with unusually high macroeconomic uncertainty and financial-market volatility, making it difficult to isolate communication effects from broader crisis-related dynamics.

2.6 Heterogeneous Monetary Transmission Across Firm Size

A substantial literature documents that monetary policy has heterogeneous effects across firms of different sizes. Gertler and Gilchrist (1994) show that smaller firms are more sensitive to monetary policy shocks due to greater dependence on external financing and bank credit.

More recent work emphasizes that firm-level differences in capital structure and financing conditions contribute to heterogeneous monetary-policy transmission (Jeenas, 2019). These findings imply that smaller firms should exhibit stronger sensitivity to monetary policy surprises than larger firms.

This literature provides the theoretical foundation for the paper's prediction that equal-weighted equity indices should respond more strongly to monetary policy shocks than value-weighted indices.

2.7 Methodological Considerations

Several methodological issues are important in monetary policy event studies.

First, the choice of event window affects identification precision. Narrow intraday windows provide cleaner identification of announcement effects, while daily-frequency returns may dilute some high-frequency information.

Second, unscheduled FOMC meetings may differ from scheduled meetings because they may convey additional information regarding economic urgency and financial conditions.

Third, the choice of monetary policy surprise measure has first-order implications for empirical results. As shown later in the paper, replacing high-frequency monetary policy shocks with simple realized rate changes substantially weakens explanatory power and statistical significance.

3. Data and Variable Construction

3.1 Monetary Policy Shocks

We use high-frequency monetary policy shocks from Acosta (2022), which extends the Gürkaynak-Sack-Swanson (GSS) decomposition through July 2022. The GSS framework decomposes monetary policy surprises into two dimensions:

Target shock: the unexpected component of the current policy-rate decision.

Path shock: revisions to expected future policy paths.

The target factor captures unexpected current-policy actions, while the path factor captures changes in market expectations regarding the future trajectory of monetary policy.

Importantly, the path factor should not be interpreted as a pure information shock in the Jarociński-Karadi (2020) sense. Rather, it reflects revisions to future policy expectations that may contain informational, forward-guidance, and expectation-management components.

The shocks are standardized to unit variance in the full Acosta sample. In our estimation sample (2006–2022), the standard deviations are 0.82 for the target shock and 0.80 for the path shock. We use the shocks as provided by Acosta without re-standardization to preserve comparability with the existing literature.

The correlation between the target and path shocks is 0.14 in our sample, confirming that the GSS decomposition successfully separates current-policy surprises from future-policy revisions.

3.2 FOMC Statement Corpus

We collect FOMC statements from January 2006 through March 2025 from the Federal Reserve’s official website. Of the 140 statements collected, 117 overlap with the Acosta monetary policy shock data and form the primary estimation sample.

The sample spans three Federal Reserve Chairs:

Ben Bernanke,
Janet Yellen,
and Jerome Powell.

Over the sample period, FOMC statements became substantially longer and more detailed, reflecting the increasing importance of communication and forward guidance in modern monetary policy implementation.

The statements are cleaned and standardized for textual analysis by removing formatting elements and non-substantive institutional text.

3.3 Sentiment Analysis

We compute two sentiment scores for each FOMC statement:

Loughran-McDonald (LM) score: The fraction of positive words minus the fraction of negative words, using the Loughran and McDonald (2011) dictionary. The LM dictionary contains 354 positive and 2,329 negative terms in its master list, of which a subset appears in FOMC text. The score is computed as:

$$LM_t = (\text{Positive words}_t - \text{Negative words}_t) / \text{Total words}_t \quad (1)$$

Central Bank (CB) score: A hawkish-dovish score computed using an expanded central bank-specific dictionary comprising 591 hawkish terms (e.g., "tightening," "inflationary pressures," "vigilant") and 222 dovish terms (e.g., "accommodative," "downside risks," "labor market slack"). The CB score is computed as:

$$CB_t = (\text{Hawkish words}_t - \text{Dovish words}_t) / \text{Total words}_t \quad (2)$$

The CB dictionary was constructed by combining terms from the existing literature (Apel and Blix, 2014; Tadle, 2022) with additional terms identified through manual review of FOMC statements. The hawkish dictionary includes terms related to inflation concerns, tightening policy, and robust economic conditions, while the dovish dictionary includes terms related to accommodation, easing, and economic weakness.

Combined score: We use the equal-weighted average of the LM and CB scores as our primary sentiment measure:

$$S_t = 0.5 \times LM_t + 0.5 \times CB_t \quad (3)$$

This follows the standard approach in the literature, but we note that the LM component exhibits a positivity bias for FOMC text (see Section 6). The equal-weighted combination is chosen for transparency and to avoid data-snooping concerns; as we show in Section 6.7, the CB score alone provides stronger results, but we prefer the combined score as our baseline specification.

3.4 Market Returns

We use CRSP daily market data obtained through WRDS as the primary source for equity returns.

The main equity measures are:

CRSP value-weighted return (vwretd): a broad market return weighted by firm market capitalization.

CRSP equal-weighted return (ewretd): an equal-weighted index that places greater weight on smaller firms.

CRSP S&P 500 return (sprtrn): the standard large-cap benchmark.

The equal-weighted index is particularly useful for examining heterogeneous monetary transmission because smaller firms tend to be more sensitive to financing conditions and external borrowing costs.

We additionally collect:

gold prices,
10-year Treasury yields,
13-week Treasury bill yields,
and the VIX

from the Federal Reserve Economic Data (FRED) database.

CRSP is used instead of alternative retail-data sources because it provides delisting-adjusted returns, which are particularly important for equal-weighted indices that contain a larger proportion of smaller firms.

3.5 Summary Statistics

Table 1 reports the summary statistics for the key variables in our estimation sample.

Table 1: Summary Statistics (N = 117, 2006-2022)

Variable	Mean	Std	Min	Max
Target shock	-0.022	0.823	-4.955	2.691
Path shock	0.003	0.795	-2.615	3.389
Combined sentiment	0.014	0.006	0.008	0.034
LM score	0.041	0.008	0.031	0.071
CB score	-0.013	0.005	-0.022	0.005
Kuttner surprise (bp)	-0.449	3.076	-20.000	9.539

Note: Target and path shocks from Acosta (2022), standardized to unit variance in the full sample (1995-2022). Combined sentiment = $0.5 \times LM + 0.5 \times CB$. Kuttner surprise in basis points.

Several features of the data deserve comment. First, the target and path shocks exhibit low correlation (0.14), confirming that the Gürkaynak-Sack-Swanson decomposition successfully separates current-policy surprises from future-policy revisions. Both shock series display substantial variation during periods of elevated monetary-policy uncertainty, including the global financial crisis and the COVID-19 period.

Second, the sentiment measures exhibit comparatively limited variation, reflecting the highly institutionalized nature of FOMC communication. The LM measure maintains a positive average level throughout the sample, consistent with the known positivity bias of general financial dictionaries when applied to central-bank communication. By contrast, the CB measure displays greater directional variation and more directly captures hawkish versus dovish policy language.

Third, the equal-weighted equity index exhibits greater sensitivity to monetary policy surprises than broader value-weighted indices, consistent with the view that smaller firms are more exposed to financing-condition channels of monetary-policy transmission.

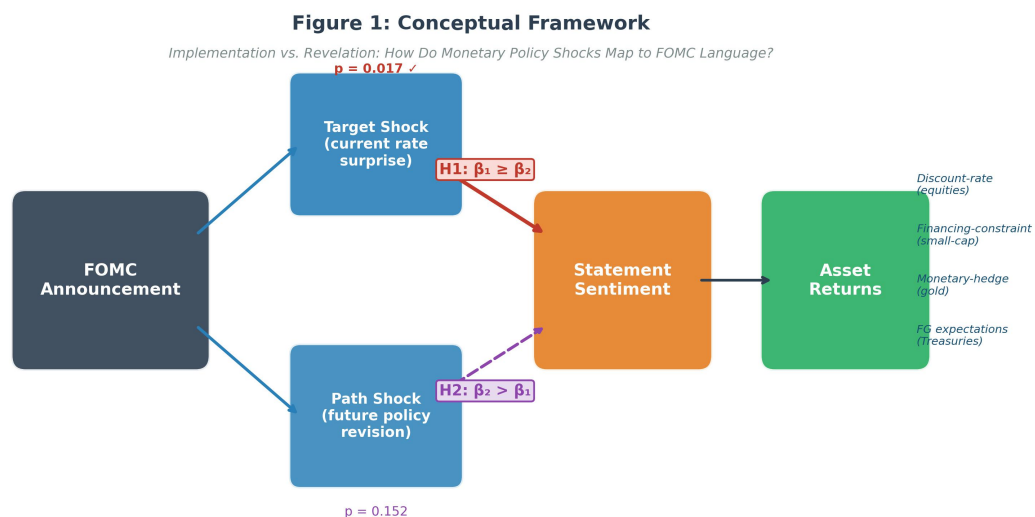
Finally, the monetary policy shock measures retain substantial variation throughout the sample period, supporting the use of high-frequency identification for studying the relationship between monetary policy surprises, statement sentiment, and cross-asset financial-market responses.

4. Empirical Methodology

Figure 1 presents the conceptual framework guiding the analysis. FOMC announcements generate two dimensions of monetary policy surprises — target shocks associated with unexpected current-rate decisions and path shocks associated with revisions to expected future policy paths. The paper examines whether these shocks predict FOMC statement sentiment and cross-asset financial-market responses.

The central empirical question is whether FOMC statement language is more closely associated with current policy implementation or future-policy revisions. If statement sentiment primarily reflects current policy implementation, target shocks should be more strongly associated with language. If statement sentiment primarily reflects forward-looking informational revelation, path shocks should play a larger role.

The cross-asset analysis further evaluates whether monetary policy shocks affect different financial assets in ways consistent with distinct transmission channels, including discount-rate effects, financing conditions, and expectations channels.



Note: Solid arrow = significant path (H1 favored); Dashed arrow = insignificant path (H2 not supported); Wald test $p = 0.90$ cannot reject equal effects.

*Figure 1: Conceptual Framework — Implementation vs. Revelation:
How Do Monetary Policy Shocks Map to FOMC Language?*

Note: Solid arrow = significant path (H1 favored); Dashed arrow = insignificant path (H2 not supported). Wald test $p = 0.90$ cannot reject equal effects.

4.1 Sentiment and Monetary Policy Shocks

We estimate the following regression to test whether FOMC statement sentiment is related to monetary policy shocks:

$$S_t = \alpha + \beta_1 \cdot \text{Target}_t + \beta_2 \cdot \text{Path}_t + \varepsilon_t \quad (4)$$

where S_t is the combined sentiment score for FOMC meeting t , Target_t is the target shock, and Path_t is the path shock.

The regression examines whether statement sentiment is more closely associated with unexpected current-policy decisions or with revisions to expected future policy paths. If FOMC language primarily reflects current policy implementation, target shocks should exhibit stronger associations with sentiment. If statement language primarily reflects forward-looking informational revelation, path shocks should play a larger role.

We conduct Wald tests of coefficient equality to evaluate whether the target and path shocks exhibit statistically different associations with statement sentiment.

Importantly, the GSS path factor should not be interpreted as a pure information shock in the Jarociński-Karadi (2020) sense. Rather, it captures revisions to expected future policy paths that may reflect informational revelation, forward-guidance commitment effects, or broader expectation-management mechanisms.

4.2 Asset Returns and Monetary Policy Shocks

To examine how monetary policy surprises transmit across financial markets, we estimate the following event-study specification for each asset class:

$$R_t = \alpha + \beta_1 \cdot \text{Target}_t + \beta_2 \cdot \text{Path}_t + \varepsilon_t \quad (5)$$

The analysis includes:

- CRSP value-weighted returns,
- CRSP equal-weighted returns,
- S&P 500 returns,
- gold returns,
- 10-year Treasury yields,
- and 13-week Treasury bill yields.

Different asset classes are interpreted as reflecting different monetary transmission channels. Broad equity-market responses are

interpreted as evidence consistent with discount-rate effects, while stronger responses among equal-weighted equities are interpreted as evidence consistent with heterogeneous financing-condition channels affecting smaller firms more strongly than larger firms.

Treasury-yield responses are interpreted as reflecting revisions to future policy expectations and forward-guidance-related effects, while gold returns provide evidence related to inflation-hedge and monetary-uncertainty channels.

The comparison between value-weighted and equal-weighted equity returns is particularly important because it allows us to examine whether smaller firms exhibit greater sensitivity to monetary policy surprises than larger firms, consistent with the financing-channel literature.

The paper does not claim to structurally identify these mechanisms. Rather, the cross-asset evidence is interpreted as providing evidence broadly consistent with distinct monetary transmission channels.

4.3 Relative Importance of Target and Path Shocks

To evaluate whether FOMC statement sentiment is more closely associated with current policy implementation or future-policy revisions, we compare the relative magnitudes of the target and path coefficients in the baseline sentiment regression.

We conduct Wald tests of coefficient equality under the null hypothesis:

$$W = (\beta_1 - \beta_2)^2 / \text{Var}(\beta_1 - \beta_2) \quad (6)$$

where $\text{Var}(\hat{\beta}_1 - \hat{\beta}_2) = \text{Var}(\hat{\beta}_1) + \text{Var}(\hat{\beta}_2) - 2\text{Cov}(\hat{\beta}_1, \hat{\beta}_2)$ is estimated using the Newey-West variance-covariance matrix. Under the null hypothesis, W follows a χ^2 distribution with 1 degree of freedom.

Failure to reject the null hypothesis implies that the two shock dimensions exhibit statistically similar associations with statement sentiment, while rejection suggests differential relationships between current-policy surprises and future-policy revisions.

The Wald statistics are computed using the Newey-West variance-covariance matrix to account for potential heteroskedasticity and autocorrelation in the regression residuals.

4.4 Forward Guidance Period Interaction

To examine whether the relationship between statement sentiment and financial-market responses differs during the forward-guidance period, we estimate the following interaction specification:

$$R_t = \alpha + \beta_1 \cdot \text{Target}_t + \beta_2 \cdot \text{Path}_t + \beta_3 \cdot S_t + \beta_4 \cdot (S_t \times \text{FG}_t) + \varepsilon_t \quad (7)$$

where FG_t is an indicator for the forward guidance period (December 2008 to December 2015, when the federal funds rate was at the zero lower bound). The interaction term evaluates whether statement sentiment exhibits stronger associations with asset returns during periods in which communication and expectation management became more central components of monetary policy implementation.

The forward-guidance period includes 48 of the 117 FOMC meetings in the estimation sample.

4.5 Estimation

All regressions are estimated using ordinary least squares with Newey-West heteroskedasticity and autocorrelation consistent (HAC) standard errors.

The use of HAC standard errors is motivated by the possibility of serial correlation and time-varying volatility in both financial-market returns and institutional communication data. As a robustness check, the paper also reports heteroskedasticity-robust standard errors. The main qualitative results remain broadly similar across specifications.

4.6 Identification Assumptions

Several identification issues should be acknowledged.

First, high-frequency monetary policy shocks may not be perfectly exogenous. The GSS framework assumes that narrow-window changes in interest-rate futures around FOMC announcements primarily reflect monetary policy surprises rather than other contemporaneous news. While this assumption is standard in the high-frequency identification literature, Bauer and Swanson (2023) show that monetary policy surprises may be partially predictable using publicly available macroeconomic information released prior to FOMC meetings.

Second, the empirical framework should be interpreted as identifying reduced-form associations rather than fully structural causal effects. Because FOMC statements and policy decisions are released simultaneously, the analysis cannot fully separate communication effects from broader policy-announcement effects.

Third, the regression specifications assume approximately linear relationships between monetary policy shocks, statement sentiment, and asset returns. Although nonlinear effects and regime dependence may exist, additional robustness analysis reported in Section 6 suggests that the main qualitative findings remain broadly similar across alternative specifications.

Finally, the relationship between monetary policy shocks and communication may vary across policy regimes and Federal Reserve Chairs. The paper therefore examines several subsample and regime-based specifications to evaluate the stability of the baseline results.

5. Results

5.1 Sentiment and Monetary Policy Shocks

Table 2 reports the baseline regression relating FOMC statement sentiment to monetary policy shocks.

The target shock exhibits a positive and statistically significant association with statement sentiment, while the path shock is positive but statistically insignificant. These results suggest that FOMC statement language is more closely associated with unexpected current-policy decisions than with revisions to expected future policy paths.

The explanatory power of the regression is modest ($R^2 = 1.57\%$), which is common in text-based financial regressions given the inherently noisy nature of institutional communication data.

A Wald test cannot reject the null hypothesis that the target and path coefficients are equal. Accordingly, the results should be interpreted cautiously. The evidence provides more support for the policy-implementation interpretation than for a strong informational-revelation interpretation, but does not conclusively eliminate informational effects.

Table 2: Sentiment and Monetary Policy Shocks

Variable	β	SE	t	p
Target shock	0.000577	0.000238	2.43	0.017
Path shock	0.000633	0.000439	1.44	0.152
Constant	0.0145	—	—	—
R^2	1.57%			
N	117			

Note: Newey-West HAC(4) standard errors. The dependent variable is the combined sentiment score ($0.5 \times LM + 0.5 \times CB$).

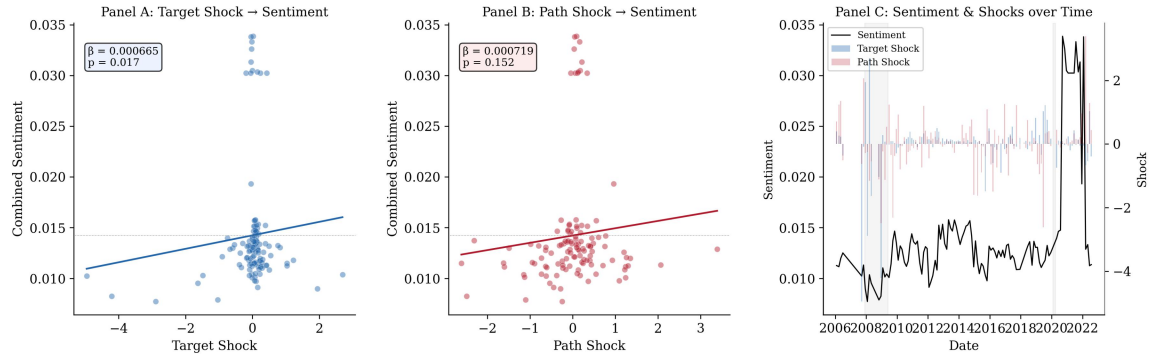


Figure 2: Sentiment vs Monetary Policy Shocks

Figure 2 illustrates the relationship between sentiment and monetary policy shocks. The target shock exhibits a visibly positive, though noisy, relationship with statement sentiment, while the path-shock relationship appears substantially weaker.

5.2 Data Source Comparison

Table 3 compares the baseline results across alternative measures of monetary policy surprises, including simple rate changes, the Kuttner (2001) surprise, and the GSS target-path decomposition.

The choice of monetary policy surprise measure substantially affects the empirical results. Replacing high-frequency shocks with simple realized rate changes weakens explanatory power and eliminates statistical significance. By contrast, the Kuttner surprise and the GSS decomposition produce economically meaningful and statistically significant relationships.

These findings highlight the importance of properly identified monetary policy surprises in communication studies. Simple realized rate changes combine anticipated and unanticipated policy components, introducing measurement noise that weakens the estimated relationship between monetary policy and communication.

Table 3: Surprise Measure Comparison (Dependent Variable: Combined Sentiment)

Surprise Measure	β (t)	p	R ²	N
Rate change	0.00103 4 (0.73)	0.72 6	1.05 %	11 7
Kuttner surprise (bp)	0.00021 2 (2.88)	0.00 4	2.14 %	11 7
GSS target shock	0.00057 7 (2.43)	0.01 7	1.57 %	11 7

Note: The GSS specification includes both target and path shocks; R^2 is for the full model. Newey-West HAC(4) standard errors.

5.3 Asset Returns and Monetary Policy Shocks

Table 4 reports cross-asset financial-market responses to monetary policy shocks.

The target shock produces economically meaningful declines across equity markets. The strongest responses occur in the CRSP equal-weighted index, followed by the value-weighted index and the S&P 500. This pattern is consistent with heterogeneous monetary transmission through financing conditions, where smaller firms are more sensitive to unexpected monetary tightening due to greater dependence on external financing and bank credit.

NASDAQ and gold also exhibit significant negative responses to target shocks, while Treasury-yield responses are comparatively weak in daily-frequency specifications.

The path shock does not produce statistically significant responses across most asset classes. While the coefficients are generally directionally consistent with theoretical expectations, the effects are substantially weaker and less precisely estimated than the target-shock responses.

Table 4: Asset Returns and Monetary Policy Shocks (CRSP Data)

Asset	β_T	t_{target}	p_{target}	β_P	p_{path}	R^2	N
CRSP VW	-0.43 5	-2.05	0.043	-0.18 6	0.443	9.10%	11 7
CRSP EW	-0.44 9	-2.53	0.013	-0.17 4	0.479	10.30 %	11 7
S&P 500	-0.39 1	-1.80	0.073	-0.17 9	0.424	7.80%	11 7
NASDAQ	-0.28 2	-2.09	0.039	-0.16 6	0.309	3.40%	11 7
Gold	-0.40 4	-2.47	0.014	-0.48 8	0.146	7.00%	11 7
10Y Treasury	0.007	0.84	0.403	-0.00 1	0.89	0.70%	11 7
13W T- bill	0.004	0.69	0.491	-0.00 3	0.737	0.70%	11 7

Note: Newey-West HAC(4) standard errors. Returns in basis points for equities and gold; percentage points for Treasury yields.

An important pattern emerges from comparing the equity indices. The equal-weighted index responds more strongly than the value-weighted index and the S&P 500, consistent with the financing-channel literature emphasizing the greater sensitivity of smaller firms to monetary-policy shocks.

Treasury-yield responses are comparatively weak, likely reflecting the use of daily-frequency rather than narrow intraday event windows. Daily returns may dilute part of the high-frequency information embedded in monetary policy announcements.

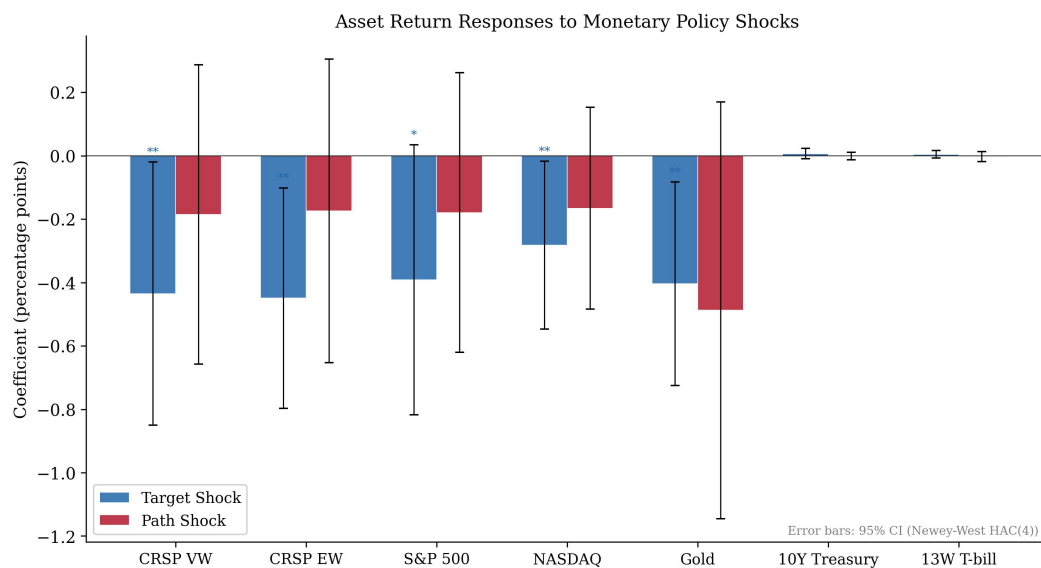


Figure 3: Asset Return Responses to Monetary Policy Shocks

Figure 3 summarizes the cross-asset responses. The target shock generates consistently negative responses across equity markets and gold, while path-shock effects remain comparatively limited.

5.4 Policy Implementation vs Informational Revelation

The central question of the paper is whether FOMC statement language primarily reflects current policy implementation or forward-looking informational revelation.

Overall, the evidence favors the policy-implementation interpretation.

Target shocks are more consistently associated with statement sentiment and equity-market responses than path shocks. Cross-asset responses are strongest among equities, particularly equal-weighted

equities, consistent with financing-condition and discount-rate channels.

By contrast, path-related effects are generally weaker and more sensitive to specification. Treasury and forward-guidance-related responses are comparatively limited in the daily-frequency regressions.

Importantly, however, the results should not be interpreted as rejecting the information channel more broadly. Federal Reserve communication extends beyond the post-meeting statement itself and includes:

- press conferences,
- Summary of Economic Projections releases,
- speeches,
- and broader expectation-management strategies.

Moreover, the GSS path factor is not a pure information shock in the Jarociński-Karadi sense. It captures revisions to expected future policy paths that may contain both informational and commitment-related components.

Accordingly, the evidence suggests that FOMC statement sentiment appears more closely tied to current policy implementation and financing conditions than to strong forward-looking informational revelation.

5.5 Mechanism Analysis: Why Is the Target Shock More Empirically Relevant?

Several interpretations may explain why target shocks are more consistently associated with statement sentiment than path shocks.

First, FOMC statements may primarily explain the rationale for the current policy decision rather than provide extensive forward guidance regarding future policy paths. Under this interpretation, current-policy surprises naturally exhibit stronger associations with statement language.

Second, forward guidance may operate through communication channels other than the statement text itself, including press conferences, speeches, and Summary of Economic Projections releases. If future-policy information is primarily transmitted through these broader communication channels, the path shock may affect market expectations without strongly affecting statement sentiment.

Third, dictionary-based sentiment measures may not fully capture subtle forms of informational communication embedded in central-bank language. Future-oriented information may be conveyed through conditional phrasing, semantic nuance, or probabilistic language that simple hawkish-dovish sentiment measures cannot fully detect.

Taken together, these interpretations suggest that the comparatively weak path-shock results may reflect both communication-channel differences and measurement limitations rather than the complete absence of informational effects.

5.6 Forward Guidance Period Interaction

We next examine whether the relationship between statement sentiment and financial-market responses differs during the zero lower bound and forward-guidance period.

Overall, we do not find robust evidence that statement sentiment becomes systematically more important during the forward-guidance period. Interaction effects are generally weak and sensitive to specification.

One NASDAQ specification produces a statistically significant interaction coefficient, but the magnitude is economically implausible and likely reflects outlier sensitivity rather than a stable forward-guidance effect.

Additional subsample analysis suggests some evidence of regime heterogeneity, particularly during rate-cut periods, where path shocks become more strongly associated with statement sentiment. However, these results are based on relatively small subsamples and should therefore be interpreted cautiously.

Overall, the full-sample results remain more robust than the regime-specific estimates.

Table 5: Forward Guidance Period Interaction

Variable	CRSP VW	NASDAQ
Target shock	−0.423	−0.284
	(0.044)	(0.034)
Path shock	−0.223	−0.089
	(0.336)	(0.622)
Sentiment	−19.99	4.54
	(0.180)	(0.771)
Sentiment × FG	−48.81	202.17
	(0.602)	(0.041)
R ²	10.0%	5.8%
N	117	117

Note: p-values in parentheses. Newey-West HAC(4) standard errors. FG = forward guidance period indicator (Dec 2008–Dec 2015). ||*|* $p < 0.01$, |*|* $p < 0.05$, |* $p < 0.1$.*

5.7 Economic Interpretation

The results carry several broader implications for monetary-policy communication.

First, the findings suggest that current-policy implementation remains the dominant driver of FOMC statement sentiment, even in an era of increasing forward guidance and expectation management.

Second, the stronger responses among equal-weighted equities provide evidence consistent with heterogeneous monetary transmission through financing conditions and balance-sheet sensitivity among smaller firms.

Third, the comparatively weak Treasury and path-shock results suggest that some forward-guidance and informational effects may operate through channels not fully captured by daily-frequency statement sentiment measures.

More broadly, the results highlight the importance of distinguishing between:

current-policy surprises,
future-policy revisions,
and broader informational communication channels

when interpreting the relationship between monetary policy and financial markets.

6. Robustness and Extensions

6.1 Alternative Standard Errors and Lag Structure

To evaluate the sensitivity of the baseline results to standard-error estimation, we re-estimate the sentiment regressions using alternative Newey-West lag structures together with White heteroskedasticity-robust standard errors.

The main qualitative findings remain stable across specifications. The target shock remains statistically significant across alternative lag choices, while the path shock remains comparatively weak and generally insignificant. The similarity of the results across standard-error estimators suggests that the baseline findings are not driven by a particular autocorrelation correction.

Overall, the results support the robustness of the central finding that target shocks exhibit stronger associations with statement sentiment than path shocks.

6.2 Alternative Monetary Policy Surprise Measures

As an alternative to the GSS target-path decomposition, we estimate the sentiment regression using the Kuttner (2001) surprise measure and simple realized federal funds rate changes.

The Kuttner surprise remains positively and significantly associated with statement sentiment, producing results broadly consistent with the baseline specification. By contrast, replacing high-frequency surprises with realized rate changes substantially weakens explanatory power and eliminates statistical significance.

These findings highlight the importance of properly identified monetary policy surprises in communication studies. Simple rate changes combine anticipated and unanticipated policy components, introducing measurement noise that weakens the estimated relationship between monetary policy and communication.

6.3 Excluding COVID

To evaluate whether the baseline results are driven by the extreme volatility associated with the COVID-19 crisis, we re-estimate the baseline specification excluding the March–June 2020 period.

The results remain broadly unchanged. The target shock remains statistically significant, while the path shock remains comparatively weak and statistically insignificant.

This robustness check suggests that the main findings are not driven by the unusually large monetary policy interventions and financial-market volatility associated with the early pandemic period.

6.4 Sentiment Dictionary Robustness

To evaluate whether the baseline results depend on the choice of sentiment measure, Table 7 compares alternative textual sentiment specifications using both FOMC statements and FOMC minutes.

Specifically, we estimate the baseline sentiment regression using:

the Loughran-McDonald (LM) dictionary alone,
the central-bank-specific (CB) dictionary alone,
and the combined sentiment measure.

We additionally examine whether sentiment extracted from FOMC minutes produces qualitatively similar results.

The results show that the central-bank-specific dictionary generally outperforms the standard LM dictionary in both explanatory power and statistical significance. This finding is consistent with the institutional nature of central-bank communication, where general financial sentiment dictionaries may fail to capture hawkish and dovish policy language effectively.

Importantly, however, the broader qualitative pattern remains stable across specifications: target shocks exhibit stronger and more consistent associations with sentiment measures than path shocks.

The minutes-based measures produce somewhat stronger path-related effects than the statement-based measures, suggesting that longer-form policy discussions may contain additional forward-looking information not fully reflected in the shorter post-meeting statements. However, these results should be interpreted cautiously because the minutes differ substantially from statements in both length and institutional purpose.

Overall, the results suggest that the paper's central conclusions are not driven solely by a particular sentiment dictionary or communication document.

Table 7: Alternative Sentiment Measures and Monetary Policy Shocks

Model	β_{Target}	p_{Target}	β_{Path}	p_{Path}	R^2
Statement ~ Shocks	0.000577 **	0.017	0.000633	0.152	1.57 %
Minutes LM ~ Shocks	0.000918 *	0.083	0.001091	0.324	3.67 %
Minutes CB ~ Shocks	0.000147	0.716	0.001423 *	0.061	5.60 %
Minutes Combined ~ Shocks	0.000532 **	0.011	0.001257 **	0.015	9.35 %
Statement ~ Shocks + Min Comb.	0.000391 *	0.062	0.000194	0.611	6.06 %

6.5 The Bauer-Swanson Critique

Bauer and Swanson (2023) argue that high-frequency monetary policy surprises may contain predictable components related to publicly available macroeconomic information released prior to FOMC meetings. This critique raises important questions regarding the exogeneity of monetary policy shocks identified using narrow event windows.

While this concern may affect the interpretation of monetary policy surprises more broadly, the main comparative findings of the paper are less likely to be driven entirely by this issue because the critique applies to both the target and path shocks. Accordingly, the relative comparison between the two shock dimensions remains informative even if the shocks are not perfectly exogenous.

More broadly, the results continue to support the view that properly identified high-frequency surprises provide substantially more informative measures of monetary policy shocks than simple realized rate changes.

6.6 Additional Robustness and Regime Variation

We additionally examine several alternative specifications, including:

post-2010 subsamples,
monetary-policy-regime subsamples,
and Federal Reserve Chair subsamples.

Some evidence of regime heterogeneity emerges, particularly during periods associated with forward guidance and rate cuts, where path-related effects become somewhat stronger. However, these results are based on comparatively small subsamples and are therefore interpreted cautiously.

Importantly, the main full-sample findings remain qualitatively stable across alternative specifications: target shocks exhibit more consistent associations with statement sentiment and equity-market responses than path shocks.

Overall, the robustness analysis supports the central conclusion that FOMC statement sentiment appears more closely tied to current policy implementation than to strong forward-looking informational revelation.

7. Conclusion and Discussion

Does FOMC statement language primarily reflect current policy implementation or forward-looking informational revelation? This paper addresses this question by combining high-frequency monetary policy shocks, textual sentiment analysis, and cross-asset financial-market responses within a unified empirical framework.

Using the Gürkaynak-Sack-Swanson target-path decomposition across 117 FOMC meetings from 2006–2022, the paper finds that target shocks are more consistently associated with FOMC statement sentiment and equity-market responses than path shocks. The target shock significantly predicts statement sentiment, while the path shock remains comparatively weak and sensitive to specification.

Cross-asset results provide additional evidence consistent with heterogeneous monetary-policy transmission. Equity-market responses are strongest among equal-weighted indices, suggesting that smaller firms are more sensitive to unexpected monetary-policy changes through financing-condition and balance-sheet channels. Treasury-yield and forward-guidance-related effects are comparatively limited in daily-frequency specifications.

Taken together, the results suggest that FOMC statement sentiment appears more closely tied to current policy implementation and financing conditions than to strong forward-looking informational revelation.

Importantly, however, the results should not be interpreted as rejecting the information channel more broadly. Federal Reserve communication extends beyond the post-meeting statement itself and includes press conferences, Summary of Economic Projections releases, speeches, and broader expectation-management strategies. Moreover, the GSS path factor is not a pure information shock in the Jarociński-Karadi (2020) sense, but rather a broader measure of revisions to expected future policy paths.

The paper also highlights several methodological considerations for future research. First, the choice of monetary policy surprise measure has first-order implications for inference. Properly identified high-frequency surprises substantially outperform simple realized rate changes in explaining communication outcomes. Second, domain-specific sentiment measures provide more informative results than general-purpose financial dictionaries when analyzing central-bank communication.

Several limitations should be acknowledged. The sentiment measures remain relatively simple dictionary-based representations of institutional communication and may not fully capture contextual or semantic nuances embedded in FOMC language. In addition, daily-frequency financial-market data may dilute some high-frequency announcement effects, particularly for Treasury yields and forward-guidance-related responses.

Future research may therefore benefit from:

- more advanced language models,
- higher-frequency market data,
- broader communication corpora,
- and structural identification approaches that separately distinguish monetary policy shocks from information shocks.

More broadly, the paper demonstrates the value of integrating textual analysis with high-frequency monetary policy identification. As central banks increasingly rely on communication and expectation management as policy instruments, understanding the relationship between monetary policy shocks and central-bank language remains an important area for both macroeconomic research and policy analysis.

Data Availability. The monetary policy shock data from Acosta (2022) are publicly available. FOMC statements are available from the

Federal Reserve website. CRSP market data are available through WRDS with an institutional subscription. The replication code and processed datasets will be made available upon publication.

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Appendix A: Expanded Central Bank Sentiment Dictionary

A.1 Hawkish Terms (591)

The expanded hawkish dictionary includes terms such as: tightening, inflationary, overheating, vigilant, hawkish, restrictive, contractionary, firming, upward pressure, price stability concerns, normalization, tapering, unwinding, balance sheet reduction, rate hike cycle, preemptive, aggressive, robust growth, labor market tightness, wage pressures, capacity constraints, elevated inflation, persistent inflation, inflation expectations rising, monetary policy normalization,

removing accommodation, less accommodative, policy firming, gradual tightening, credible commitment to price stability, inflation risks skewed to the upside, upside risks to inflation, diminishing slack, approaching mandate-consistent levels, well-positioned to respond, data-dependent tightening, measured pace of tightening, appropriate degree of policy restraint.

A.2 Dovish Terms (222)

The expanded dovish dictionary includes terms such as: accommodative, easing, dovish, expansionary, stimulative, supportive, downside risks, labor market slack, subdued inflation, below target, disinflationary, persistent slack, economic headwinds, financial stability concerns, cautious approach, patient, data-dependent easing, extended period, considerable time, balanced risks, appropriate accommodation, maintaining accommodation, insufficient progress, disappointing, weakening, softening, contraction, recession risks, downside risks to growth, inflation running below, inflation expectations declining, need for continued support, premature tightening, risk of stalling, fragile recovery, uneven progress, transitory factors, temporary headwinds.

Appendix B: Data Sources Summary

Variable	Source	Frequen cy	Covera ge
Target/P ath shocks	Acosta (2022)	Per meeting	1995-- 2022
FOMC statements	Fed website	Per meeting	2006-- 2025
CRSP VW/EW/S&P returns	WRDS (crsp.dsi)	Daily	1990-- 2024
Gold price	FRED (GOLDAMGBD228NL BM)	Daily	1968-- 2025
10Y Treasury yield	FRED (DGS10)	Daily	1962-- 2025
13W T- bill yield	FRED (DGS3MO)	Daily	1981-- 2025
VIX	FRED (VIXCLS)	Daily	1990-- 2025
Fed Funds Rate	FRED (DFF)	Daily	1954-- 2025

Financial stocks	WRDS (crsp.dsf)	Daily	2020-- 2025
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Appendix C: Additional Robustness

C.1 Regime-Specific Results

Regime	N	R ²	β_T (p)	β_P (p)
Rate hike	1 7	10.2 %	−0.0005 54 (0.013)	−0.0001 45 (0.298)
Rate cut	1 1	43.1 %	0.00018 8 (0.089)	0.00083 7 (0.000)
Unchang ed	8 9	2.0 %	0.00063 8 (0.616)	0.00137 9 (0.079)

During rate cut meetings, although path-related effects appear stronger during rate-cut periods, these estimates are based on very small subsamples and should therefore be interpreted cautiously. $p < 0.001$ and the R^2 is 43.1%, suggesting that forward guidance language is most responsive to the path shock when the Fed is easing. During rate hike meetings, the target shock is more empirically relevant ($p = 0.013$). When rates are unchanged, neither shock is significant at 5%.

C.2 Sentiment Distribution

Statistic	Combined	LM Score	CB Score
Mean	0.014	0.041	−0.013
Std	0.006	0.008	0.005
Min	0.008	0.031	−0.022
Max	0.034	0.071	0.005
% Negative	0%	0%	92.3%
% Positive	100%	100%	1.7%

The LM score is always positive for FOMC statements (min = 0.031), because FOMC statements use more positive than negative words regardless of policy stance. The CB component has substantial sign variation (92.3% negative, 1.7% positive, 6.0% zero), reflecting the predominantly dovish language in our sample period. The equal-weighted combination dilutes this signal.

C.3 Newey-West Lag Sensitivity (H1 Regression)

Lag	β_T (t)	p_target	β_P (t)	p_path	R ²
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1	0.000577 (2.78)	0.006	0.000633 (1.64)	0.1	1.57%
2	0.000577 (2.61)	0.01	0.000633 (1.55)	0.123	1.57%
4	0.000577 (2.43)	0.017	0.000633 (1.44)	0.152	1.57%
6	0.000577 (2.29)	0.024	0.000633 (1.45)	0.149	1.57%

C.4 Data Source Comparison (S&P 500)

Data Source	β_T	t_target	p_target	R²
CRSP (sprtrn)	-0.391	-1.80	0.073	0.078
yfinance (^GSPC)	-0.259	-2.19	0.03	0.029

The yfinance coefficient is 34% smaller in magnitude than the CRSP coefficient, reflecting the absence of delisting adjustments in the yfinance data.